

Signal and Image Processing

Signal Processing

IDENTIFICATION

CODE : IF-3-TSI
ECTS : 2.0

HOURS

Lectures : 15.0 h
Seminars : 6.0 h
Laboratory : 8.0 h
Project : 0.0 h
Teacher-student
contact : 29.0 h
Personal work : 10.0 h
Total : 39.0 h

ASSESSMENT METHOD

Practical work [8 hours on 2 weeks]
and exam [1 hour]

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

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AIMS

This course aims to present the general aspects of signal processing.

CONTENT

It starts by a mathematical survey of Fourier's series, as well as a cursory presentation of the notion of distribution in order to introduce the Dirac. Students will then study the transform of Fourier, the convolution operator, notions of transfer functions, impulse responses and complex spectrums. The second part of the class focus on digital signals and sampling and its consequences, of Fourier's transform and filtering. A lab. work[« TP »] is associated with this course and will allow students to use the Fourier transform to visualize digital signal spectrums and construct different filtering.

BIBLIOGRAPHY

- [1] COTTET F., Traitement des signaux et acquisition de données, DUNOD.
- [2] GASQUET C., WITOMSKI P., Analyse de Fourier et applications, MASSON.
- [3] KUNT M., Traitement numérique des signaux, DUNOD.
- [4] JAIN A.K., Fundamentals of digital image processing, PRENTICE-HALL INT. ED.

PRE-REQUISITE

Complex numbers, integral calculus